



Aun Muhammad

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ABOUT ME

As a recent graduate with a Bachelor's degree in Materials Science and Engineering, I possess a comprehensive understanding of material properties, behaviors, and applications across various industries. My academic journey has honed my ability to analyze, interpret, and optimize complex data, allowing me to provide insightful solutions to intricate engineering challenges. I am deeply committed to leveraging cutting-edge trends and technological advancements to drive innovation and efficiency in materials design and applications. With a steadfast positive mindset and an eagerness to learn, I thrive in dynamic environments that demand adaptive thinking and a proactive approach. I am eager to contribute my skills in both collaborative team settings and independent projects, aiming to foster sustainable solutions that push the boundaries of engineering possibilities.

EDUCATION AND TRAINING

09/2020 – 09/2024 Islamabad, Pakistan

BACHELORS OF ENGINEERING IN MATERIALS SCIENCE AND ENGINEERING Institute of Space Technology

Core Subjects

- Extraction of materials
- Industrial chemistry
- Mechanics of materials
- Materials thermodynamics
- Crystal structures and analysis
- Polymers
- Metal and alloys
- Ceramic materials
- Nano materials
- Material characterization
- Corrosion and materials prevention
- Manufacturing and casting
- Welding and joining
- Advance composite materials
- Health safety and environment
- Professional practice
- Engineering management

Website www.ist.edu.pk | **Field of study** Engineering, manufacturing and construction

2018 – 2020 Rawalpindi, Pakistan

INTERMEDIATE PRE-ENGINEERING FG Sir Syed, The Mall

Website fgssc.edu.pk

2016 – 2018 Rawalpindi, Pakistan

MATRICULATION IN SCIENCE Army Public School and College System

Website www.apsasc.com

WORK EXPERIENCE

01/2025 – CURRENT Islamabad, Pakistan

INTERN NATIONAL CENTER FOR FAILURE ANALYSIS

- Responsibilities related to Root Cause Failure Analysis.
- Still imaging, Conventional Metallography, Etchant compositions, Replica Metallography.
- Optical Microscopy, Stereo Microscopy, SEM, SEM-EDS, XRD Analysis.
- Mechanical Testing (Tensile Test, Compression Test, Impact Test)
- Report writing.

31/01/2024 – 08/03/2024 Islamabad, Pakistan

INTERNATIONAL CENTER FOR FAILURE ANALYSIS

- Conducted detailed Root Cause Failure Analysis (RCFA) to investigate mechanical, material, and operational failures in various industrial components.
- Utilized advanced analytical techniques, such as metallography, scanning electron microscopy (SEM), and spectroscopy, to diagnose failure modes.
- Collaborated with a multidisciplinary team to assess failure mechanisms and recommend corrective actions.
- Prepared comprehensive failure analysis reports, including findings, root cause identification, and actionable solutions for clients.
- Gained hands-on experience in testing and evaluating material properties under stress and environmental conditions.
- Contributed to enhancing industrial reliability and safety standards by implementing effective failure prevention strategies.

07/2023 – 07/2024 Rawalpindi, Pakistan

INTERN ADVANCED COMPOSITES LAB, IST

- Assisted in the development of piezo resistive sensors utilizing the centrifugal spinning technique for advanced proprioceptive applications.
- Conducted detailed mechanical testing to evaluate sensor performance under varying stress and strain conditions.
- Characterized sensor materials using techniques such as microscopy, spectroscopy, and electrical property analysis.
- Collaborated with a multidisciplinary research team to optimize sensor design and material properties.
- Contributed to the development of innovative sensing technologies with potential applications in robotics and wearable devices.

● **LANGUAGE SKILLS**

Mother tongue(s): **URDU**

Other language(s): **ENGLISH**

● **PROJECTS**

10/2023 – 07/2024

Recycling of Polymers using Centrifugal Spinning

- Successfully recycled PET polymer waste and upcycled it into high-quality crystalline fibers using advanced processing techniques.
- Conducted comprehensive testing and characterization of crystalline fibers, evaluating mechanical, thermal, and structural properties.
- Designed an eco-friendly process to transform waste PET into value-added products, demonstrating environmental and economic impact.
- Secured selection for business incubation based on the project's potential for commercial scalability and sustainable development.
- Developed expertise in polymer processing, fiber production, and advanced material testing methods.
- Thesis and Presentations

10/2023 – 07/2024

Smart Sensing and Structural Health Monitoring of Advanced Composites

- Assisted in the fabrication of piezo-resistive sensors for structural health monitoring.
- Conducted comprehensive testing and characterization of the sensors
- Developed expertise in polymer processing, fiber production, and advanced material testing methods.

● **RESEARCH INTERESTS**

Inspection and Testing of Materials

High Temperature Materials

Advance Composites

Coating and Design Modifications

Additive Manufacturing

Soft Robotics

Sustainable Materials

● **EQUIPMENT BASED SKILLS**

SEM Analysis

XRD Analysis

SMAW Machine

GMAW Machine

Centrifugal Spinning

Metallography

Electric Furnace

Hot Press

Hardness Testing

● **MANAGEMENT AND LEADERSHIP SKILLS**

Project Management

Root Cause Failure Analysis Of Materials

Adaptability

Quality Control

Digital Literacy

Communication Skills

Write here the description...

Ethical Literacy

● **DIGITAL SKILLS**

MS OFFICE | AUTOCAD | MAT LAB | VESTA | data processing and analysis software: Origin (advanced) | Zoom

● **RECOMMENDATIONS**

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